Module-2

IoT Sensing and Actuation

Introduction, Sensors, Sensor Characteristics, Sensorial Deviations, Sensing Types, Sensing Considerations, Actuators, Actuator Types, Actuator Characteristics.

Textbook 1: Chapter 5 – 5.1 to 5.9

Introduction

- A major IoT applications involves sensing in one form or the other.
- Sensing forms the first step.
- Incidentally, actuation forms the final step in the whole operation of IoT application deployment.
- Sensing and actuation is based on the process of transduction.
- Transduction is the process of energy conversion from one form to another.
- A transducer is a physical means of enabling transduction.
- Transducers take energy in any form (for which it is designed)—electrical, mechanical, chemical, light, sound, and others—and convert it into another, which may be electrical, mechanical, chemical, light, sound, and others.
- For example, in a public announcement (PA) system, a microphone (input device) converts sound waves into electrical signals, which is amplified by an amplifier system (a process). Finally, a loudspeaker (output device) outputs this into audible sounds by converting the amplified electrical signals back.

Table 5.1 Basic outline of the differences between transducers, sensors, and actuators

Parameters	Transducers	Sensors	Actuators
Definition	Converts energy from one form to another.	Converts various forms of energy into electrical signals.	Converts electrical signals into various forms of energy, typically mechanical energy.
Domain	Can be used to represent a sensor as well as an actuator.	It is an input transducer.	It is an output transducer.
Function	Can work as a sensor or an actuator but not simultaneously.	Used for quantifying environmental stimuli into signals.	Used for converting signals into proportional mechanical or electrical outputs.
Examples	Any sensor or actuator	Humidity sensors, Temperature sensors, Anemometers (measures flow velocity), Manometers (measures fluid pressure), Accelerometers (measures the acceleration of a body), Gas sensors (measures concentration of specific gas or gases), and others	Motors (convert electrical energy to rotary motion), Force heads (which impose a force), Pumps (which convert rotary motion of shafts into either a pressure or a fluid velocity).

Sensors

- A sensor is a device that detects and responds to some type of input from the physical environment.
- The input can be light, heat, motion, moisture, pressure or any number of other environmental phenomena.
- They generate responses to external stimuli or physical phenomenon through characterization of the input functions (which are these external stimuli) and their conversion into typically electrical signals.
- For example, heat is converted to electrical signals in a temperature sensor, or atmospheric pressure is converted to electrical signals in a barometer.

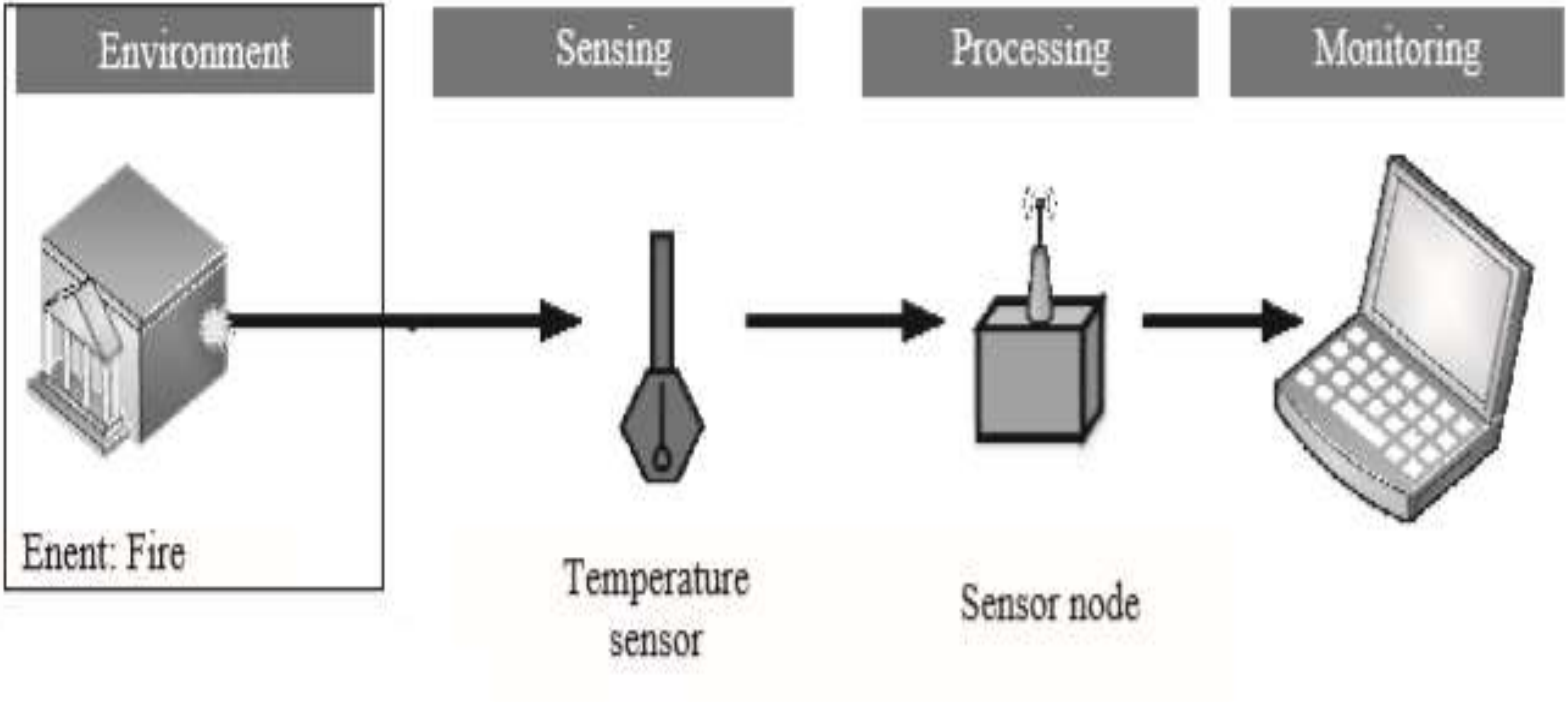


Figure 5.1 The outline of a simple sensing operation

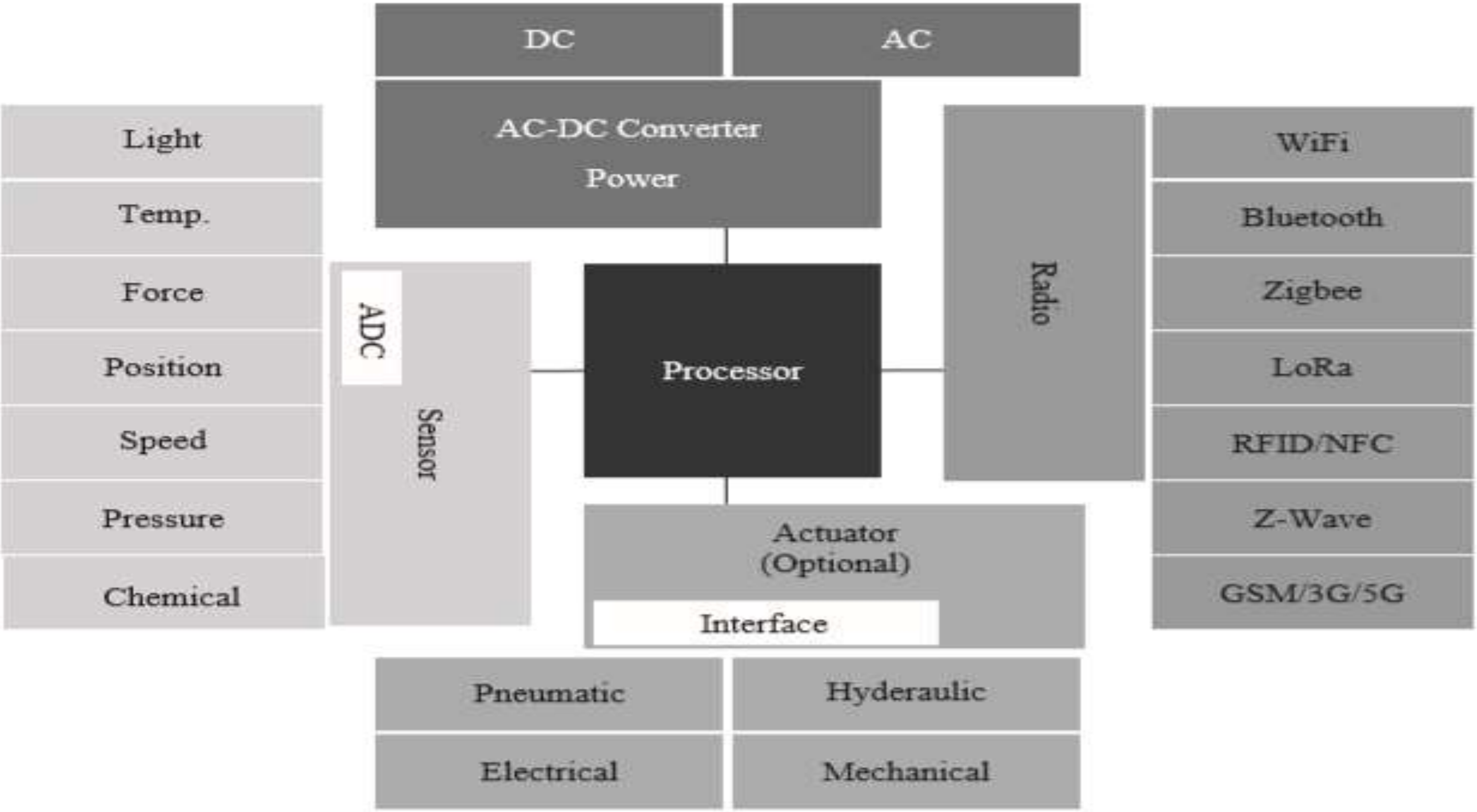


Figure 5.2 The functional blocks of a typical sensor node in IoT

(a) Camera sensor (b) Color sensor (c) Compass and barometer (d) Current sensor (e) Digital temperature and humidity sensor

(f) Flame sensor (g) Gas sensor (h) Infrared sensor (i) Rainfall sensor (j) Ultrasonic distance measurement sensor

Figure 5.3 Some common commercially available sensors used for IoT-based sensing applications

- A sensor node is made up of a combination of

1. sensor/sensors,
2. processor unit,
3. radio unit, and
4. power unit.

- The nodes are capable of sensing the environment they are set to measure and communicate the information to other sensor nodes or a remote server.
- Typically, a sensor node should have low-power requirements and be wireless. This enables them to be deployed in a vast range of scenarios and environments without the constant need for changing their power sources or managing wires.
- The wireless nature of sensor nodes would also allow them to be freely relocatable and deployed in large numbers without bothering about managing wires.

- The various sensors can be classified based on:

1) power requirements,
2) sensor output, and
3) property to be measured.

- **Power Requirements:** Depending on the requirements of power, sensors can be of two types.

(i) Active: Do not require an external circuitry or mechanism to provide it with power. It directly responds to the external stimuli from its ambient environment and converts it into an output signal.

For example, a photodiode converts light into electrical impulses.

(ii) Passive: Require an external mechanism to power them up. The sensed properties are modulated with the sensor's inherent characteristics to generate patterns in the output of the sensor.

For example, a thermistor's resistance can be detected by applying voltage difference across it or passing a current through it.

Sensor Output

- The output of a sensor helps in deciding the additional components to be integrated with an IoT node or system.
- Sensors are broadly divided into two types, depending on the type of output generated from the sensors.
- **Analog**
- **Digital**

Digital Sensors vs Analog

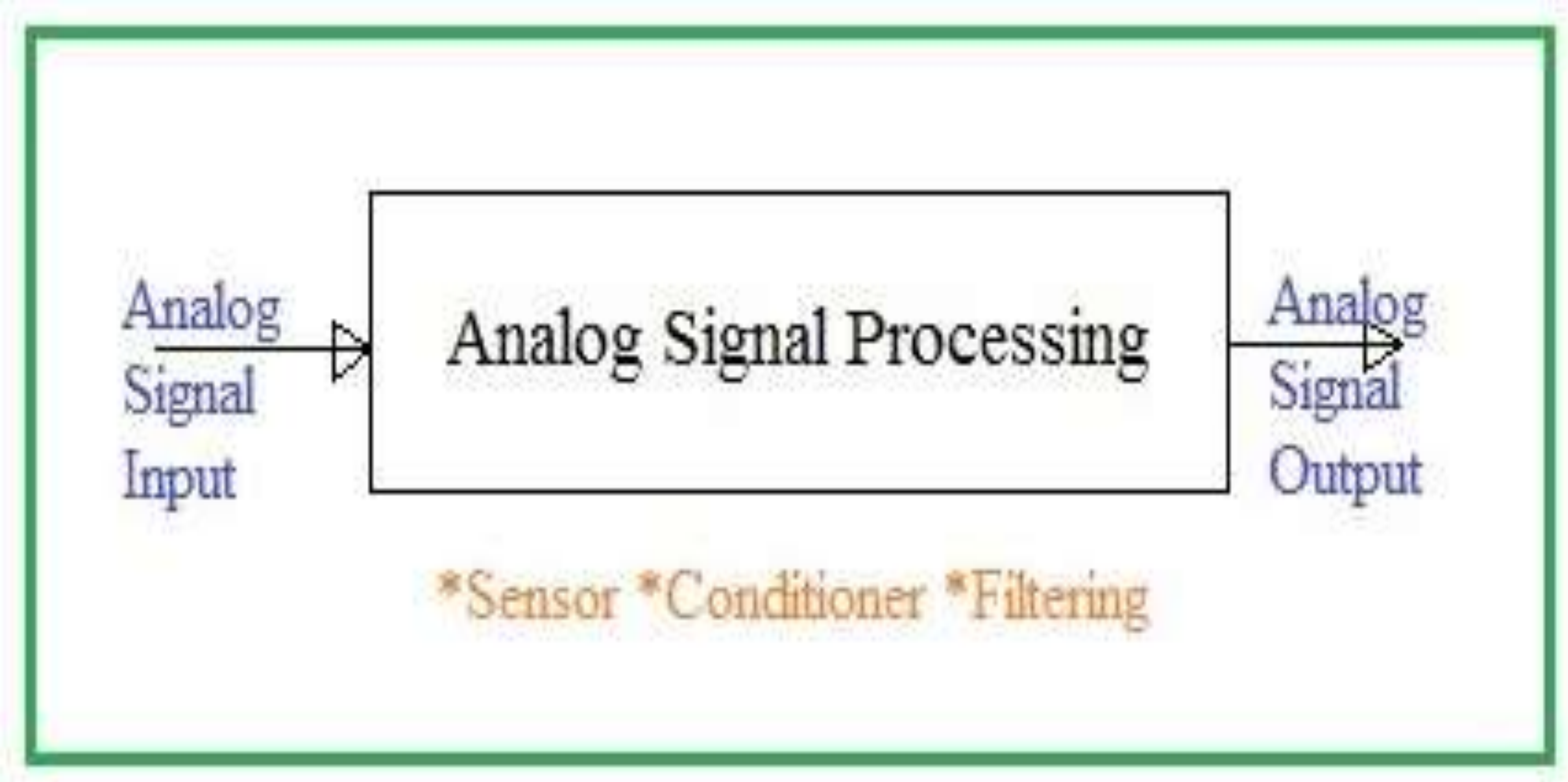
Digital

- Less prone to noise
- One line can transmit multiple sensors
- Need to interpret the 1's and 0's
- Need to make sure clocks a synced up
- An agreed upon language must be set up in order to communicate

Analog

- Can be directly measured
- Usually simpler setups
- No compatibility issues
- More prone to noise
- Needs an ADC for computer interpretation

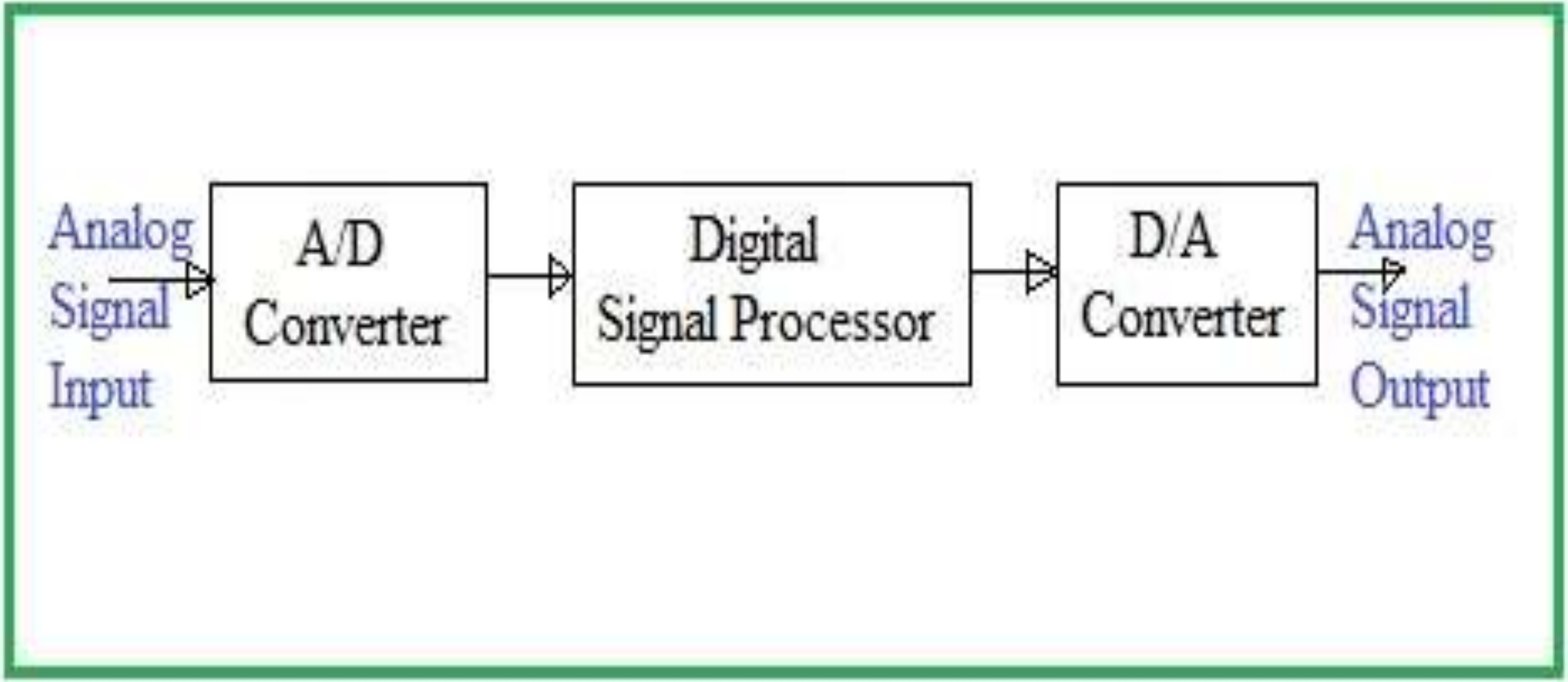
Analog sensor



Analog sensor

- It produces continuous output signal or voltage which is proportional to quantity to be measured.
- It produces analog output.
- Quantities such as temperature, speed, displacement, pressure, strain etc. are analog quantities as they are continuous in nature
- Example: Temperature of liquid can be measured using thermometer or thermocouple which continuously responds to changes in temperature as liquid is heated up or cooled down.

Digital sensor



Digital sensor

- It produces discrete digital output signal or voltage which is digital representation of quantity to be measured.
- It produces binary output in the form of ones (1s) and zeros (0s).
- Digital sensors overcome limitations of analog sensors counterpart.
- They are used in various applications such as waste water, water and other industrial processes.
- Digital sensor consists of sensor itself, cable and a transmitter.
- Examples: Measurements such as pH level, conductivity, dissolved oxygen, ammonium, nitrate etc. are conducted using digital sensors.

Measured Property

- The property of the environment being measured by the sensors can be crucial in deciding the number of sensors in an IoT implementation.

- Depending on the properties to be measured, sensors can be of two types.

- **(i) Scalar:**
- **(ii) Vector**

Scalar sensor

- The sensor which produce output signal/voltage which is proportional to magnitude of quantity to be measured is known as scalar sensor.

- Temperature of a room is measured using scalar sensor (i.e. thermometer or thermocouple) which measures temperature changes irrespective of sensor orientation.

- Eg: Temperature sensor, Color sensor, pressure sensor, strain sensor etc.

Vector sensor

- The sensor which produce output signal/voltage which is proportional to magnitude, direction as well as orientation of quantity being measured is known as Vector sensor.
- Acceleration of body is measured using accelerometer which measures component of acceleration of body with respect to x, y, z coordinate axes.
- Examples: Sound sensor, image sensor, velocity sensor, acceleration sensor etc.

Sensor Characteristics

- Sensors can be characterized by their ability to sense the phenomenon based on the following three fundamental properties.
- **Sensor Resolution**: The smallest change in the measurable quantity that a sensor can detect is referred to as the resolution of a sensor.
- For digital sensors, the smallest change in the digital output that the sensor is capable of quantifying is its sensor resolution.
- The more the resolution of a sensor, the more accurate is the precision.
- A sensor’s accuracy does not depend upon its resolution.
- For example, a temperature sensor A can detect up to 0.5, where as another sensor B can detect up to 0.25. Therefore, the resolution of sensor B is higher than the resolution of sensor A.
- **Sensor Accuracy**: The accuracy of a sensor is the ability of that sensor to measure the environment of a system as close to its true measure as possible.
- For example, a weight sensor detects the weight of a 100 kg mass as 99.98 kg. We can say that this sensor is 99.98% accurate, with an error rate of ±0.02%.

- **Sensor Precision**:

- Precision can be described as the reproducibility of measurements.
- Only upon multiple repetitions, the sensor is found to have the same error rate, can it be deemed as highly precise.
- For example, consider if the same weight sensor described earlier reports measurements of 98.28 kg, 100.34 kg, and 101.11 kg upon three repeat measurements for a mass of actual weight of 100 kg.
- Here, the sensor precision is not deemed high because of significant variations in the temporal measurements for the same object under the same conditions.

Sensorial Deviations

A) Full-scale range: In the event of a sensor's output signal going beyond its designed maximum and minimum capacity for measurement, the sensor output is truncated to their maximum or minimum values, which are also the sensor's limits. The measurement range between a sensor's characterized minimum and maximum values is also referred to as the full-scale range of that sensor.

B) Offset error or bias: Under real conditions, the sensitivity of a sensor may differ from the value specified for that sensor leading to sensitivity error. This deviation is mostly attributed to sensor fabrication errors and its calibration. If the output of a sensor differs from the actual value to be measured by a constant, the sensor is said to have an offset error or bias. For example, while measuring an actual temperature of $0\,^{\circ}C$, a temperature sensor outputs $1.1\,^{\circ}C$ every time. In this case, the sensor is said to have an offset error or bias of $1.1\,^{\circ}C$.

C) Non-linearity: some sensors have a non-linear behavior. If a sensor’s transfer function (TF) deviates from a straight line transfer function, it is referred to as its non-linearity.

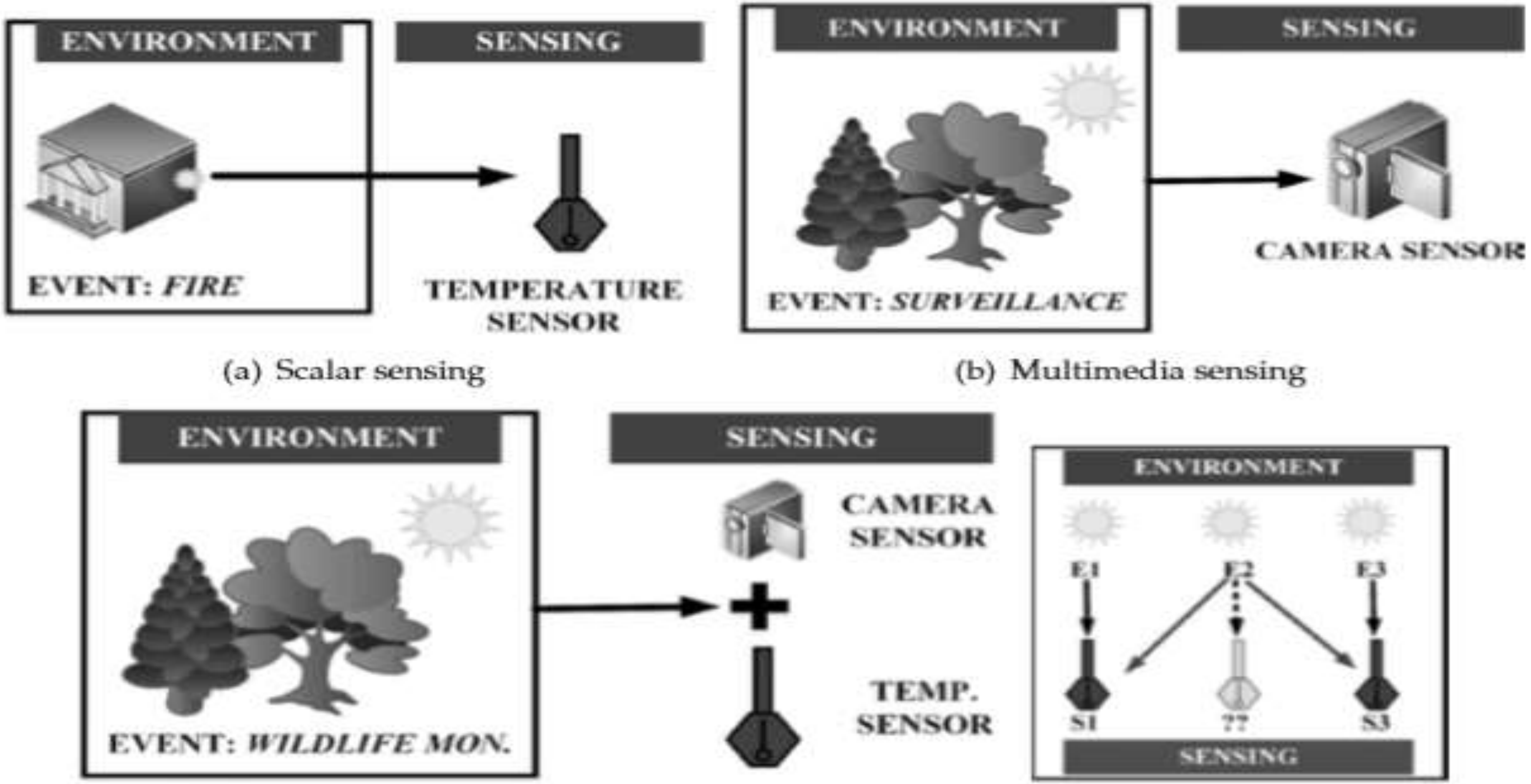
D) Drift: If the output signal of a sensor changes slowly and independently of the measured property, this behavior of the sensor’s output is termed as drift. Physical changes in the sensor or its material may result in long-term drift, which can span over months or years. Noise is a temporally varying random deviation of signals.
In contrast, if a sensor’s output varies/deviates due to deviations in the sensor’s previous input values, it is referred to as hysteresis error.

E) Quantization error : Focusing on digital sensors, if the digital output of a sensor is an approximation of the measured property, it induces quantization error. This error can be defined as the difference between the actual analog signal and its closest digital approximation during the sampling stage of the analog to digital conversion

Sensing Types

- Sensing can be broadly divided into four different categories based on the nature of the environment being sensed and the physical sensors.

1. Scalar sensing,
2. Multi media sensing,
3. Hybrid sensing and
4. Virtual sensing.



(a) Scalar sensing (b) Multimedia sensing

(c) Hybrid sensing (d) Virtual sensing

Figure 5.4 The different sensing types commonly encountered in IoT

Scalar sensing

- Scalar sensing encompasses the sensing of features that can be quantified simply by measuring changes in the amplitude of the measured values with respect to time.

- Quantities such as ambient temperature, current, atmospheric pressure, rainfall, light, humidity, flux, and others are considered as scalar values as they normally do not have a directional or spatial(3D) property assigned with them.

- Simply measuring the changes in their values with passing time provides enough information about these quantities.

Multimedia sensing

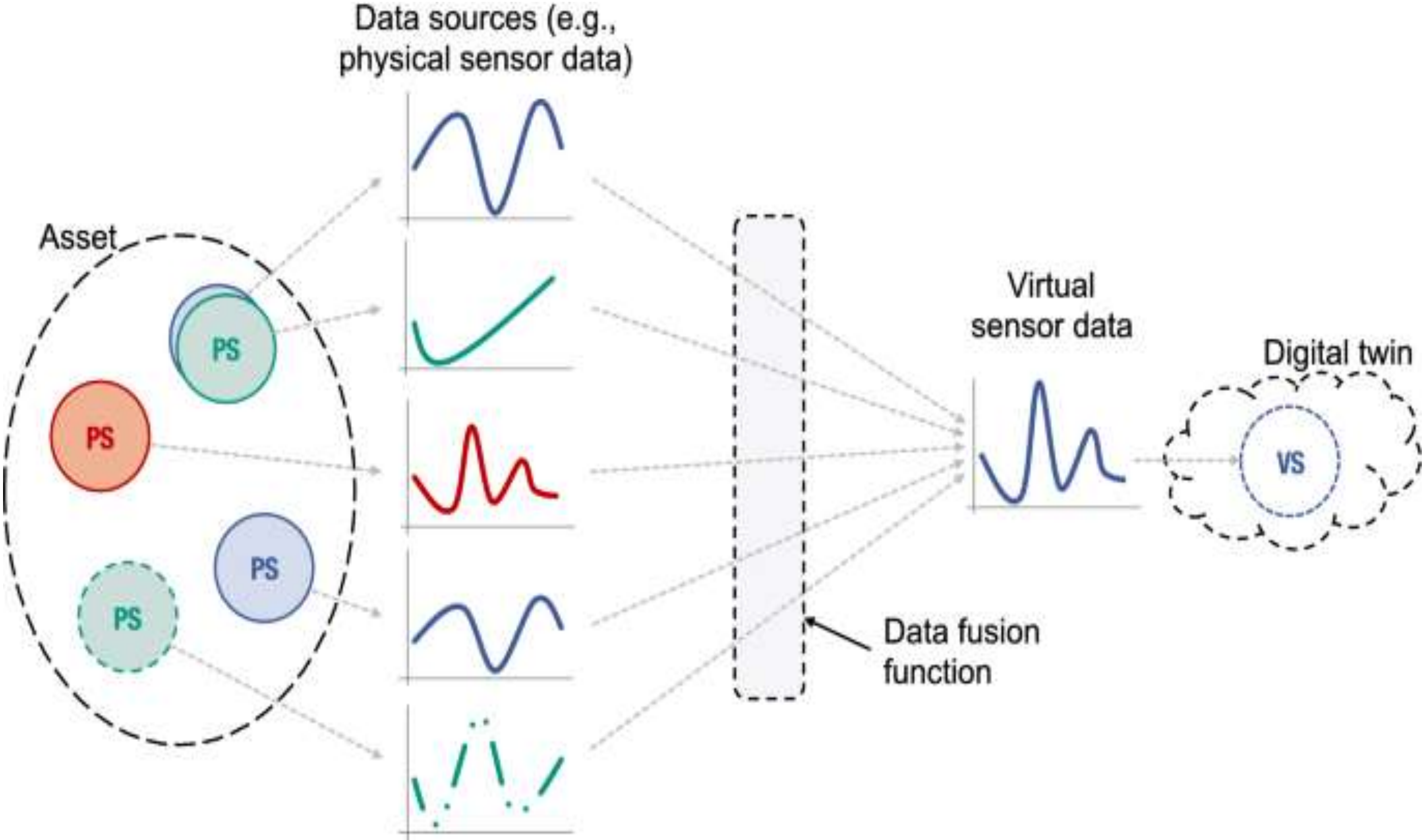
- Multimedia sensing encompasses the multiple parameters like amplitude and time (direction and magnitude).
- Vector quantities like image, direction, speed, sound, energy.

Hybrid sensing

- Hybrid sensing encompasses both scalar as well as multimedia sensing at the same time is referred to as hybrid sensing.
- Example: Camera and temperature sensor are collectively used to detect and confirm forest fire during wildlife monitoring.

Virtual sensing

- Virtual sensors represent a software layer that provides indirect measurements of a process variable or an abstract condition based on data gathered.
- It learns to understand the relationships between the different variables, and observes readings from the different instruments.
- Virtual sensing encompasses very dense and large-scale deployment of sensor nodes spread over a large area for monitoring of parameters.
- Example: Agriculture domain. Here, the parameters being measured, such as soil moisture, soil temperature, and water.
- Figure 5.4(d) shows an example of virtual sensing.
- Two temperature sensors S1 and S3 monitor three near by events E1, E2, and E3 (fires). The event E2 does not have a dedicated sensor for monitoring it; however, through the superposition of readings from sensors S1 and S3, the presence of fire in E2 is inferred.


Data sources (e.g.,
physical sensor data)
Asset
PS
PS
PS
PS
Virtual
sensor data
Digital twin
VS
Data fusion
function

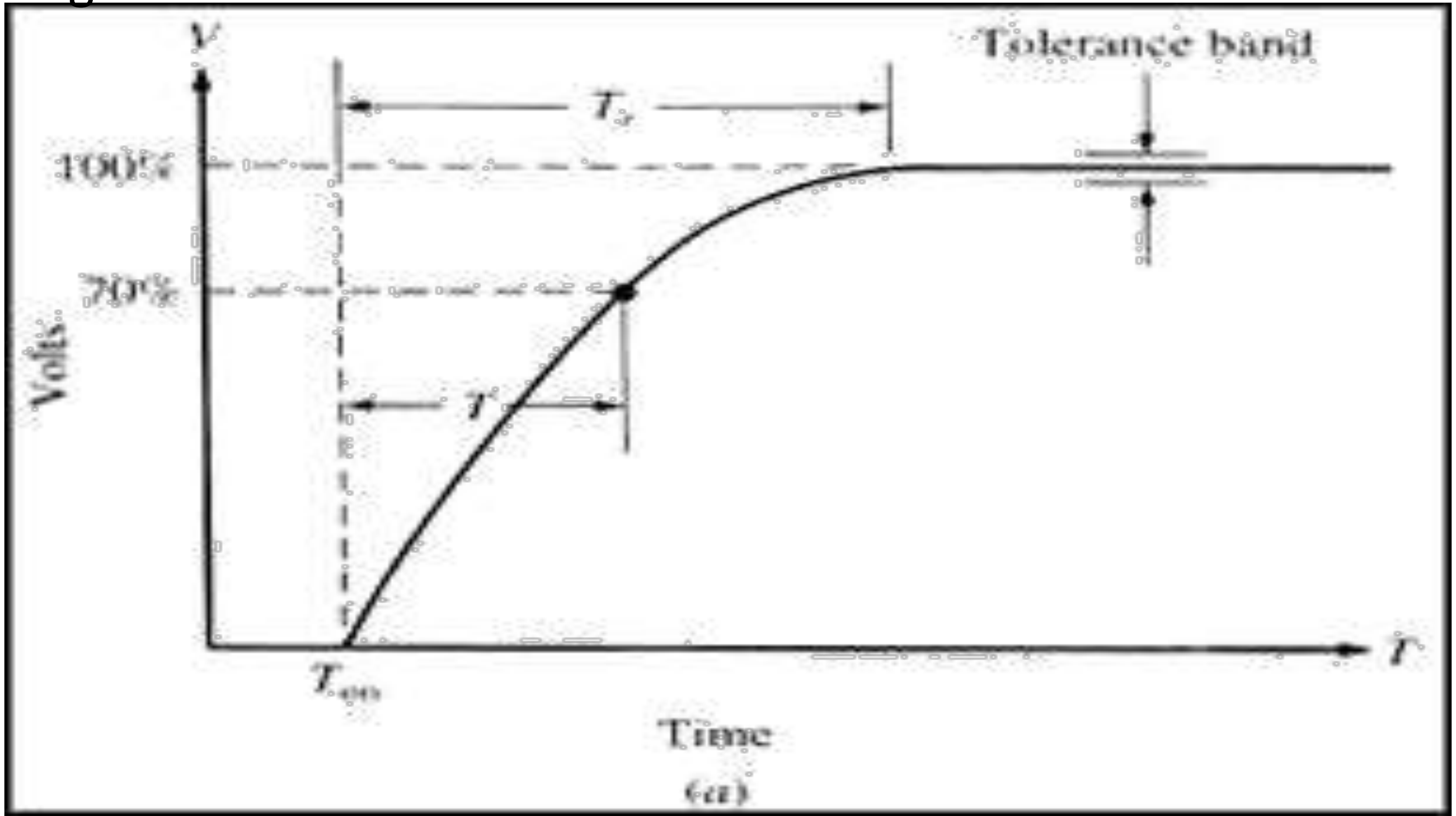
Sensing Considerations

➢ The major factors influence the choice of sensors in IoT-based sensing solutions:

1. Sensing range,
2. Accuracy and Precision,
3. Energy
4. Device size.

Sensing Range:

The range of the sensor is the maximum and minimum values of applied parameter that can be measured. For example, a given pressure sensor may have a range of -400 to +400 mm Hg.



Sensing Range:

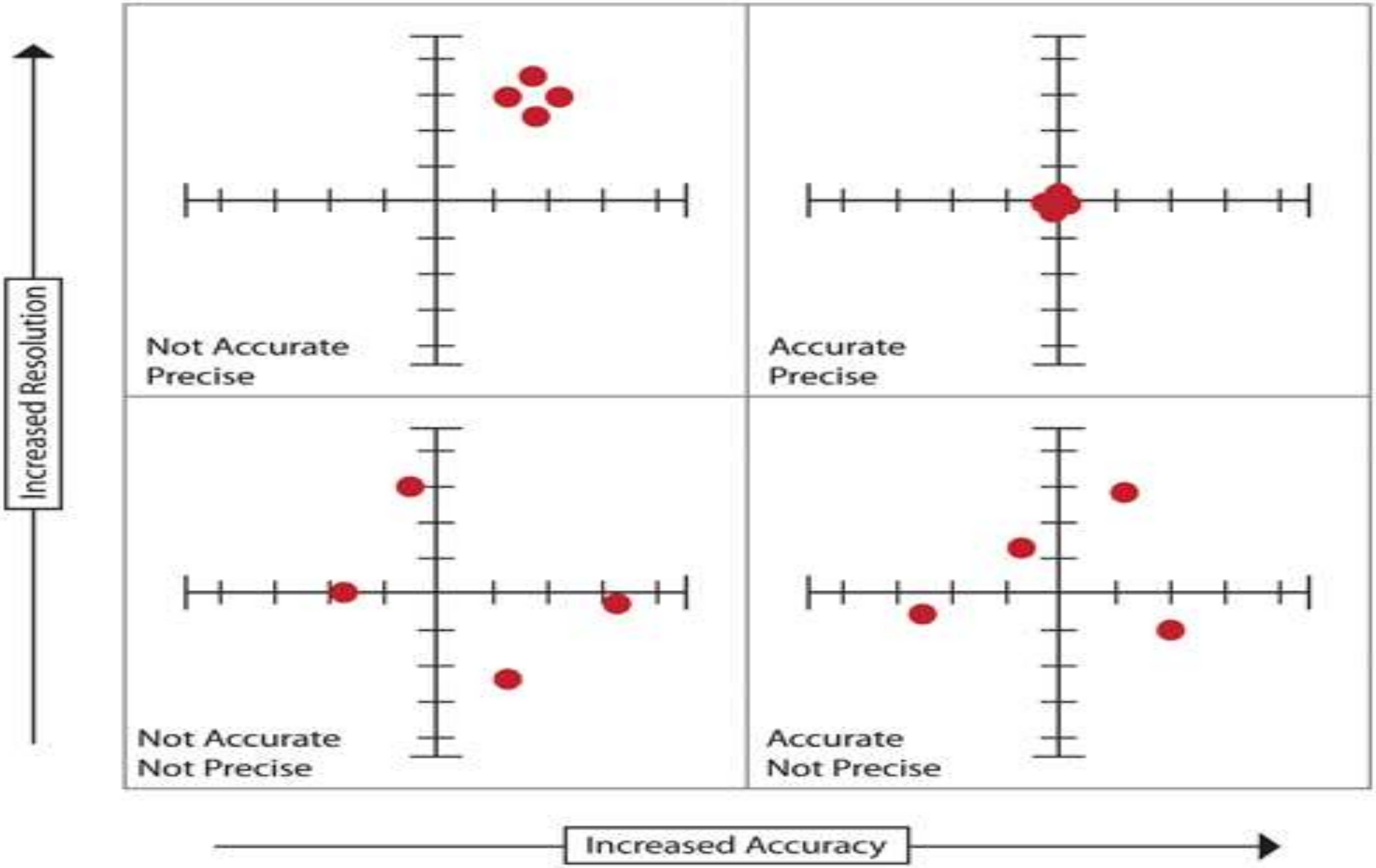
- ➢The sensing range of a sensor node defines the detection reliability of that node.
- ➢Sensing range in deployments include
- • Fixed k-coverage
- • Dynamic k-coverage.
- ➢**Fixed k-coverage** : It requires a large number of sensor nodes, the sensing range of some of which may also overlap.
- ➢**Dynamic k-coverage**: Incorporates mobile sensor nodes post detection of an event, which, however, is a costly solution and may not be deployable in all operational areas and terrains.
- ➢Additionally, the sensing range of a sensor may also be used to signify the upper and lower bounds of a sensor's measurement range.
- ➢For example a camera has a sensing range varying between tens of meters to hundreds of meters.
- ➢As the complexity of the sensor and its sensing range goes up, its cost significantly increases.

Accuracy and Precision

- The accuracy and precision of measurements provided by a sensor are critical in deciding the operations of specific functional processes.

- **Accuracy** refers to how close a measurement is to the true or accepted value.

- **Precision** refers to how close measurements of the same item are to each other.
- Accuracy **provides a high level of quality and precision**


Increased Resolution
Not Accurate
Precise
Accurate
Precise
Not Accurate
Not Precise
Accurate
Not Precise
Increased Accuracy

Energy

- The energy consumed by a sensing solution is crucial to determine the lifetime of that solution and the estimated cost of its deployment.

- If the sensor or the sensor node is so energy inefficient that it requires replenishment of its energy sources quite frequently, the effort in maintaining the solution and its cost goes up; whereas its deployment feasibility goes down.

Device Size

- Modern-day IoT applications have a wide penetration in all domains of life.
- Most of the applications of IoT require sensing solutions which are so small that they do not hinder any of the regular activities that were possible before the sensor node deployment was carried out.
- Larger the size of a sensor node, larger is the obstruction caused by it, higher is the cost and energy requirements, and lesser is its demand for the bulk of the IoT applications.
- Consider a simple human activity detector. If the detection unit is too large to be carried or too bulky to cause hindrance to regular normal movements, the demand for this solution would be low.
- It is because of this that the onset of wearables took off so strongly.
- The wearable sensors are highly energy-efficient, small in size, and almost part of the wearer's regular wardrobe.